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DUNMAN HIGH SCHOOL, SENIOR HIGH
PRELIMINARY EXAMINATION 2008
Higher 1

MATHEMATICS

8863/01

Paper 1

11 September 2008

3 hours

Additional Materials: Answer Paper
 Graph paper
 List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Name, Class and Index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, **attach the question paper to the front of your answer script.**

The number of marks is given in brackets [] at the end of each question or part question.

For teacher's use:

Section A	Marks
1	/ 3
2	/ 5
3	/ 7
4	/ 6
5	/ 9
6	/ 10

Section B	Marks
7	/ 6
8	/ 9
9	/ 10
10	/ 10
11	/ 11
12	/ 14
Total	

Section A: Pure Mathematics [40 marks]

- 1 Find, by differentiation, the range of values of x for which the function $x^3 - 3x^2 - 24x + 10$ decreases as x increases. [3]

- 2 Differentiate $\ln\left(\frac{1}{1-3x}\right)$ with respect to x . [2]

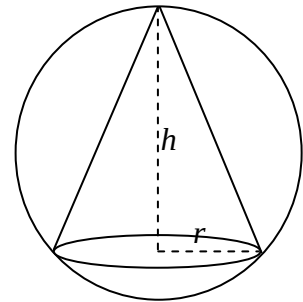
The gradient function of a curve is given by $\frac{1}{1-3x} + 2$. Find the equation of the curve if the curve passes through the point $(0, 3)$. [3]

- 3 The equation of a curve is given by $y = e^{2x}$.

- (i) Show that the equation of the tangent to the curve at $x = 2$ is $y = e^4(2x - 3)$. [3]
- (ii) Find the exact value of the area of the region bounded by the curve, the tangent at $x = 2$ and both the axes. [4]

- 4 A right circular cone with base radius r cm and height h cm is inscribed in a sphere of radius 4 cm as shown in the diagram.

Express r in terms of h and, hence, show that the volume of the cone, $V \text{ cm}^3$, is given by $V = \frac{1}{3}\pi h^2(8 - h)$. [2]



[The volume of a cone is $\frac{1}{3}(\text{base area}) \times (\text{height})$.]

Using a non-calculator method, find the exact value of h when the volume of the cone is a maximum and calculate this maximum value. [4]

- 5 The functions f and g are defined by

$$f : x \mapsto (x+3)^2 - 1, \quad x > -3,$$

$$g : x \mapsto x - 2, \quad x \in \mathbb{R}.$$

- (i) Show that the composite function fg does not exist. [2]
- (ii) Sketch the graph of $y = f(x)$ and explain clearly why the inverse function f^{-1} exists. [2]
- (iii) Find a similar expression for f^{-1} . [3]
- (iv) On the same diagram as the graph of $y = f(x)$, sketch the graph of $y = f^{-1}(x)$, and state the relationship between the two graphs. [2]
- 6 (a) (i) State the vertical and horizontal asymptotes of $y = \frac{1}{x+1}$. [2]
- (ii) Sketch the graphs of $y = x^2 - 3$ and $y = \frac{1}{x+1}$ on the same diagram, showing all the asymptotes, intercepts and stationary points. [2]
- (iii) Hence, solve the inequality $x^2 \geq \frac{1}{x+1} + 3$. [2]
- (b) (i) Sketch, marking clearly any intersections with the axes, the graph of $y = 4 \cos\left(\frac{x}{2}\right)$ for $0 \leq x \leq \pi$. [1]
- (ii) Find all the possible values of m if $4 \cos\left(\frac{x}{2}\right) = mx$ has a root between 1 and 2, where m is an integer. Show your method clearly. [3]

Section B: Statistics [60 marks]

- 7 (i) Give one similarity and one difference between stratified sampling and quota sampling. [2]
- (ii) A market researcher has been commissioned by a firm to conduct a survey of consumer spending at a local supermarket. The firm wants various age groups to be represented.
- (a) Explain why the methods of systematic sampling and stratified sampling would not be suitable. [2]
- (b) Describe how a quota sample of size 100 could be taken from the population of customers. [2]

- 8** The probability that a certain type of cactus seed will germinate is p . In a long-term study, 1500 of these seeds were planted, of which 600 germinated.

(i) Write down an estimate of p . [1]

Suzy plants 24 such seeds at the beginning of the growing season in three batches, each batch consisting of 8 seeds.

(ii) Assuming that the seeds germinate independently, use the value of p found in (i) to find

(a) the probability that at least 12 seeds germinate, [3]

(b) the most probable number of seeds that will germinate in a batch of 8, [2]

(c) the probability that only one batch produces more than 4 germinating seeds. [3]

- 9** In each round of a quiz, a contestant is asked a sequence of up to 5 questions. The round stops when the contestant has answered *two* questions in a row wrongly or has attempted all five questions. One point is awarded for each correct answer.

As an example, Mary gets the first question wrong, the second question correct, and the third and fourth questions wrong; the round then stops and she has scored 1 point. Mary's probability of getting any particular question correct is 0.6, independently of other questions. She takes part in one round of the quiz.

(a) Find the probability that Mary scores exactly one point. [3]

(b) Find the probability that Mary scores exactly two points given that she is asked fewer than 5 questions. [4]

Let X denote the number of points a contestant scores in a quiz.

Given that $E(X) = 2.54$, and $\text{Var}(X) = 2.46$. Estimate the probability that the average score of 70 contestants is more than 2.5. [3]

- 10 (a)** For a sample of bivariate data, (x_i, y_i) , $i = 1, 2, \dots, 5$, the value of the linear product moment correlation coefficient is -1 . Illustrate how this data would appear on a scatter diagram and draw the regression lines of x on y and y on x on the scatter diagram. [2]

- (b)** A random sample of 10 students is taken from those sitting examinations, and their Literature marks and History marks are denoted by x and y respectively. These ten pairs of values of x and y are used to obtain the following equations of the regression lines of y on x and of x on y respectively,

$$y = 2.883 + 0.9853x, \quad x = 9.622 + 0.8131y.$$

- (i)** Use the equations of the regression lines to deduce the product moment correlation coefficient of the data and state what this indicates about the relation between x and y over the data range. [2]

- (ii)** Nine of the ten pairs of data are given in the table below.

Literature mark (x)	54	42	80	38	72	61	53	78	56
History mark (y)	65	50	85	40	68	65	52	82	42

By finding the mean point $(\bar{x}, \bar{y})$ of the sample, find the tenth pair of integer values x and y . [3]

- (iii)** Calculate an estimate for the Literature mark of a student who scored 70 in the History examination, using the appropriate regression line. [2]

- (iv)** Give a reason why it might be unwise to use either of the regression lines to estimate the History mark of a student who scored 30 in the Literature examination. [1]

- 11** At a bottling plant, wine bottles are filled automatically by a machine. The bottles are meant to hold 500 ml of wine each, on average. A random sample of 60 bottles is examined and the quantity x ml of wine in each bottle is determined. The results are summarized by $\sum (x - 500) = 132$ and $\sum (x - 500)^2 = 4552$.

- (i) Find unbiased estimates of the population mean and variance. [3]
- (ii) Test, at the 5% level of significance, whether the bottles are being filled correctly and explain, in the context of the question, the meaning of 'at the 5% significance level'. [5]
- (iii) In a 1-tail test of the null hypothesis that the mean quantity of wine in each bottle is 500 ml, the alternative hypothesis is accepted. State the alternative hypothesis, and find an inequality satisfied by the significance level of the test. [3]

- 12** The weekly takings at three Supermart stores are modeled as independent normally distributed random variables with means and standard deviations as shown in the table, in dollars.

	Mean	Standard deviation
Supermart A	7200	250
Supermart B	9500	320
Supermart C	5000	150

- (i) Find the probability that the weekly takings at Supermart B will be at least twice those at Supermart C. [4]
- (ii) Find the probability that that in a 4-week period, the total takings at Supermart A will be less than \$30 000. [3]
- (iii) The Supermart parent company receives a weekly levy consisting of 11% of the weekly takings at Supermart A, 15% of those at Supermart B and 5% of those at Supermart C. Show that the probability that this levy exceeds \$2500 in any given week is 0.277. Hence find the probability that this levy exceeds \$2500 more than 18 times in a year by using a suitable approximation. (Assume that there are 52 weeks in a year.) [7]